The exact scattering integral S_{nl} in wim_simple: formulation, discretisation and validation

This note documents the exact evaluation of the non-linear transfer S_{nl} implemented in the sub-package snl_exact of wim_simple ("snonlin_backend" = 'exact'). It complements the model documentation, which contains only a summary, with the underlying conventions, the derivation chain of the coupling coefficient, the geometry of the resonance loci, the discretisation on the spectral grid, the discrete conservation properties, and the validation against independent implementations. Only deep water is considered.

Conventions and the kinetic equation

The prognostic variable of wim_simple is the wave action spectral density $A(\boldsymbol{k})$ per d^2k, related to the surface elevation spectrum $F(\boldsymbol{k})$, with $\langle\zeta^2\rangle = \int F\, d\boldsymbol{k}$, and to the energy density $E = \sigma A$ by

$$A \;=\; \frac{g}{\sigma}\,F \quad , \quad \sigma(\boldsymbol{k}) = \sqrt{g|\boldsymbol{k}|} \tag{1}$$

In the model, spectra are carried as functions of cyclic frequency ν and direction θ; with $k = (2\pi\nu)^2/g$ and $dk/d\nu = 8\pi^2\nu/g$ the conversion used by the exact backend is

$$A(k,\theta) \;=\; \frac{g\,E(\nu,\theta)}{\sigma\,k\,dk/d\nu} \tag{2}$$

where $E(\nu,\theta)$ is the variance density per Hz and radian.

The evolution of A by resonant four-wave interactions is the kinetic equation of Hasselmann (1962), written here in the Zakharov form

$$S_{nl}(\boldsymbol{k}_1) = 4\pi \int T^2_{1234}\,\delta(\boldsymbol{k}_1+\boldsymbol{k}_2-\boldsymbol{k}_3-\boldsymbol{k}_4)\,\delta(\sigma_1+\sigma_2-\sigma_3-\sigma_4)\;\mathcal{B}\;d\boldsymbol{k}_2\,d\boldsymbol{k}_3\,d\boldsymbol{k}_4 \tag{3}$$

with the density bracket

$$\mathcal{B} \;=\; A_3A_4(A_1+A_2) - A_1A_2(A_3+A_4) \tag{4}$$

Two properties of $\mathcal{B}$ are used repeatedly below: it vanishes identically for constant A, and it vanishes on the resonance manifold for the Rayleigh–Jeans equilibrium $A \propto 1/\sigma$, since

$$\mathcal{B}|_{A=c/\sigma} \;=\; c^3\,\frac{(\sigma_1+\sigma_2)-(\sigma_3+\sigma_4)}{\sigma_1\sigma_2\sigma_3\sigma_4} \;=\; 0 \tag{5}$$

Both properties are preserved by the discretisation (see below) and provide sharp correctness tests.

The kernel T_{1234} in Eq. (3) is the canonical four-wave kernel of the Zakharov equation

$$i\,\partial_t b_1 \;=\; \int T_{1234}\,b_2^*b_3b_4\,\delta(\boldsymbol{k}_1+\boldsymbol{k}_2-\boldsymbol{k}_3-\boldsymbol{k}_4)\,d\boldsymbol{k}_2\,d\boldsymbol{k}_3\,d\boldsymbol{k}_4 \tag{6}$$

in the Fourier convention without 2π prefactors, for which $\langle b(\boldsymbol{k}) b^*(\boldsymbol{k}')\rangle = A(\boldsymbol{k})\,\delta(\boldsymbol{k}-\boldsymbol{k}')$ holds with the physical action density Eq. (1) directly, and $T(\boldsymbol{k},\boldsymbol{k},\boldsymbol{k},\boldsymbol{k}) = k^3$. In the convention of Krasitskii (1994), with a factor $1/2\pi$ per transform, the same kernel reads $T^K = T/(2\pi)^2$ and $T^K(\boldsymbol{k},\boldsymbol{k},\boldsymbol{k},\boldsymbol{k}) = k^3/4\pi^2$; the collision prefactor 4π in Eq. (3) is form-invariant under this change of convention. The combination

$$G \;=\; 4\pi\, T^2 \;=\; 4\pi(2\pi)^4 \left(T^K\right)^2 \tag{7}$$

is the deep-water coupling coefficient of Webb (1978).

Derivation and verification of the kernel

The kernel is not transcribed from the literature but derived from the water-wave Hamiltonian, truncated at fourth order in wave steepness,

$$H_3 \;=\; \frac{1}{2}\int \zeta\left((\boldsymbol{\nabla}\psi)^2 - (G_0\psi)^2\right) d\boldsymbol{x} \tag{8}$$

$$H_4 \;=\; \frac{1}{2}\int \zeta\,(G_0\psi)\left(G_0(\zeta\, G_0\psi) + \zeta\,\nabla^2\psi\right) d\boldsymbol{x} \tag{9}$$

with the surface potential ψ, the flat-surface Dirichlet–Neumann operator G_0 (symbol $|\boldsymbol{k}|$ in deep water), and normal variables defined by

$$\hat{\zeta}(\boldsymbol{k}) = \sqrt{\frac{k}{2\sigma}}\left(a_{\boldsymbol{k}} + a^*_{-\boldsymbol{k}}\right) \quad , \quad \hat{\psi}(\boldsymbol{k}) = -i\sqrt{\frac{\sigma}{2k}}\left(a_{\boldsymbol{k}} - a^*_{-\boldsymbol{k}}\right) \tag{10}$$

(here written in the 2π-free convention). Both truncated Hamiltonians were verified exactly against the analytic kinetic energy of a prescribed harmonic flow, with residuals scaling as the neglected orders ε^3, ε^4, ε^5. The canonical (normal-form) transformation then removes the non-resonant cubic terms; on the four-wave resonance manifold the two classes of induced quartic contributions coincide pairwise, which is exploited by keeping both with weight $1/4$ so that the $(12)\leftrightarrow(34)$ symmetry of T is manifest also off the manifold.

The resulting kernel satisfies, at machine precision, the on-shell Hermiticity and index symmetries, rotation invariance, cubic homogeneity $T(\lambda\boldsymbol{k}_i) = \lambda^3 T(\boldsymbol{k}_i)$, and independence of g. It reproduces the Stokes second-harmonic and mean-flow response, the bichromatic super- and sub-harmonic amplitudes, and the two-wave degenerate-quartet interaction coefficient of Longuet-Higgins/Leblanc (as compiled by Stuhlmeier, 2024) to 10^{-6} (limited by the finite-amplitude offsets of those references). Finally, the identity

$$\text{xc_webb} \;=\; 64\pi^5 g^2 \left(T^K\right)^2 \;=\; g^2\, G \tag{11}$$

connecting Eq. (7) to the coupling routine of the exact-NL code of van Vledder (mod_xnl4v5, as distributed with WAVEWATCH III and SWAN) was verified to a relative accuracy of $3\cdot 10^{-13}$ on resonant quadruplets, so that the absolute normalisation of Eq. (3) is anchored to an independent production code at the level of the coupling coefficient.

WRT reduction and the resonance loci

Following Webb (1978) and Tracy and Resio (1982), the wavenumber delta function in Eq. (3) eliminates $\boldsymbol{k}_2$, and for fixed $(\boldsymbol{k}_1, \boldsymbol{k}_3)$ the frequency delta function reduces the remaining $\boldsymbol{k}_2$-plane integral to a line integral along the resonance locus

$$z(\boldsymbol{k}_2) \;=\; q + \sigma(\boldsymbol{k}_2) - \sigma(\boldsymbol{k}_2 + \boldsymbol{P}) \;=\; 0 \quad , \quad \boldsymbol{P} = \boldsymbol{k}_1 - \boldsymbol{k}_3 \;,\; q = \sigma_1 - \sigma_3 \tag{12}$$

with $\boldsymbol{k}_4 = \boldsymbol{k}_2 + \boldsymbol{P}$, so that

$$S_{nl}(\boldsymbol{k}_1) \;=\; \int d\boldsymbol{k}_3 \oint_{z=0} \frac{ds}{|\boldsymbol{\nabla}_{\boldsymbol{k}_2} z|}\, G\, \mathcal{B} \tag{13}$$

Note that $z(\boldsymbol{k}_3) = 0$ identically: the trivial quadruplet $\boldsymbol{k}_2 = \boldsymbol{k}_3$, $\boldsymbol{k}_4 = \boldsymbol{k}_1$ lies on every locus, with bounded G; the vanishing of the integrand there resides entirely in $\mathcal{B}$.

In deep water the loci form a one-parameter family: scaling lengths by $|\boldsymbol{P}|$ and frequencies by $\sqrt{g|\boldsymbol{P}|}$, the locus shape depends only on

$$\hat{q} = \frac{q}{\sqrt{g|\boldsymbol{P}|}} \in (0, 1) \tag{14}$$

(the labels $1, 3$ are chosen such that $q > 0$; $\hat{q} < 1$ holds for every pair since $(\sqrt{k_1} - \sqrt{k_3})^2 < k_1 - k_3 \le |\boldsymbol{k}_1 - \boldsymbol{k}_3|$). In the normalised frame with $\hat{\boldsymbol{P}} = (1, 0)$ the two intersections of the locus with the symmetry axis are analytic,

$$x_+ = \left(\frac{1 - \hat{q}^2}{2\hat{q}} \right)^2 \quad , \quad x_- = -\left(\frac{\sqrt{2 - \hat{q}^2} - \hat{q}}{2} \right)^2 \tag{15}$$

which provide exact anchors for the tracing: the closed curve is obtained by radial continuation about the interior midpoint of Eq. (15), with the derivative of the radius along the curve computed analytically from the gradient,

$$r'(\theta) = -\frac{\partial z / \partial \theta}{\partial z / \partial r} \quad , \quad ds = \sqrt{r^2 + r'^2}\, d\theta \tag{16}$$

The curve is then resampled to equal arc length, and the periodic trapezoid rule yields spectrally accurate quadrature weights $w_s = \Delta s / |\boldsymbol{\nabla} z|$. For $q = 0$ (equal frequencies, $\boldsymbol{k}_1 \neq \boldsymbol{k}_3$) the locus degenerates to the perpendicular bisector $\{\boldsymbol{k}_2 : \boldsymbol{k}_2 \cdot \boldsymbol{P} = -|\boldsymbol{P}|^2/2\}$; these same-frequency interactions are unbounded in extent but their line integral converges through the decay of the spectrum, and they are included with a separate straight-line quadrature. The dimensional line measure follows from the scale invariance,

$$\frac{ds}{|\boldsymbol{\nabla} z|} \;=\; \frac{|\boldsymbol{P}|^{3/2}}{\sqrt{g}}\, \frac{d\hat{s}}{|\hat{\boldsymbol{\nabla}} \hat{z}|} \tag{17}$$

and the coupling scales as $G \propto |\boldsymbol{k}|^6$ by the homogeneity of T. Combining the scalings of Eq. (13), the transfer obeys the deep-water self-similarity

$$S_{nl}\left(\boldsymbol{k}; \lambda A(\cdot/\mu)\right) \;=\; \lambda^3 \mu^{19/2}\, S_{nl}\left(\boldsymbol{k}/\mu; A\right) \tag{18}$$

which is reproduced by the discretisation exactly on the geometric grid (see below).

Discretisation

The spectral grid of wim_simple is geometric in frequency and uniform in direction, hence geometric in k with uniform spacing h in $\ln k$. Two consequences are exploited. First, $\hat{q}$ depends only on the ratio k_3/k_1 and the direction difference, so the complete locus geometry, the coupling values, the quadrature weights and the interpolation stencils reduce to a table over configuration classes $(\Delta i, \Delta j)$ of frequency-index and direction-index offsets: each configuration is traced once, exactly, and applied to every wavenumber pair of that class by index shifts. There is no interpolation in $\hat{q}$; the discrete grid realises finitely many configurations, and all of them are solved. Second, the stencils are shift-invariant: magnitudes enter as offsets in $\ln k$ relative to i_1, directions as offsets relative to j_1, which renders the discrete operator exactly equivariant under rotations of the spectrum by grid directions and under shifts along the geometric k-axis — the discrete form of Eq. (18).

Action densities at the off-grid locus points $\boldsymbol{k}_2, \boldsymbol{k}_4$ are obtained by bilinear interpolation in $(\ln k, \theta)$. Locus points within a small fraction of $|\boldsymbol{P}|$ of the trivial point $\boldsymbol{k}_2 = \boldsymbol{k}_3$ are removed: there the exact bracket Eq. (4) vanishes but its interpolant does not; the computed transfer is insensitive to the filter width at the percent level, which is verified as part of the test battery. Above the discrete frequency range the spectrum is extended by a parametric $E \propto \nu^{-5}$ tail over a number of additional geometric rings before evaluation (following WRT practice in WAVEWATCH III), and the transfer is restricted to the original band afterwards; quadruplets with any member beyond the extended range are discarded entirely (strict truncation). Strict truncation is what preserves the equilibria: every retained quadruplet carries all four action values, so $\mathcal{B} \equiv 0$ termwise for constant A, and Eq. (5) holds up to interpolation error for $A \propto 1/\sigma$.

With cell areas $\Delta_i = k_i\, hk_i\, \Delta\theta$ the discrete form of Eq. (13) for one quadruplet instance $(i_1 j_1, i_3 j_3, s)$ carries the measure

$$\mu_q \;=\; \Delta_3\, w_s\, G_s \tag{19}$$

and two assemblies of the same integral are provided. In the *direct* mode the rate $\mu_q \mathcal{B}$ is accumulated at $\boldsymbol{k}_1$ only — the discrete analogue of Eq. (13) and of other WRT implementations. In the *conservative* mode each instance deposits the total-action increments

$$(\delta\mathcal{A}_1, \delta\mathcal{A}_2, \delta\mathcal{A}_3, \delta\mathcal{A}_4) \;=\; (+1, +1, -1, -1)\; \frac{\Delta_1 \mu_q \mathcal{B}}{4} \tag{20}$$

distributed over the grid through the same bilinear stencils used for the gather. Dividing the deposit at the interpolated legs by the effective cell area $\bar{\Delta}$, interpolated in $\ln k$ with the same weights, conserves the deposited action exactly, since the stencil weights w_t and the θ-independent areas satisfy $\sum_t w_t \Delta_{c(t)} = \bar{\Delta}$ identically. The factor $1/4$ in Eq. (20) averages the four role-estimators of the enumeration over ordered pairs; the total action budget of the discrete transfer, including the action exchanged with the parametric tail, then closes at machine precision by construction. The two modes are independent discretisations of Eq. (13); their difference is $O(h)$, decreases under grid refinement, and serves as a built-in convergence diagnostic. Energy and momentum are conserved by the continuous

quadruplets and hence to interpolation accuracy by the discrete transfer; the residuals are reported as diagnostics.

For the semi-implicit time stepping the diagonal functional derivative is accumulated alongside,

$$\frac{\partial S_{nl}(\boldsymbol{k}_1)}{\partial A_1} \;=\; \sum_q \mu_q \left(A_3 A_4 - A_2(A_3 + A_4)\right) \tag{21}$$

which is invariant under the conversion between A and E on the diagonal and can be used directly in the update $E \leftarrow E + S\,\Delta t/(1 - \Delta t\,\partial S/\partial E)$.

Validation

The implementation is validated layer by layer; the quoted accuracies refer to the test battery shipped with the sub-package (test grids of 26–44 frequencies, 24–36 directions).

- *Kernel*: Stokes second harmonic and bichromatic super/sub harmonics exact; degenerate two-wave interaction coefficient to $4 \cdot 10^{-6}$; on-shell symmetries, rotation, homogeneity and g-independence to 10^{-13}; identity Eq. (11) with the WAVEWATCH III coupling routine to $3 \cdot 10^{-13}$.

- *Loci*: residuals $|z|$ at returned points $4 \cdot 10^{-14}$; crossings Eq. (15) exact; resonance closure of dimensional quadruplets 10^{-14} for both signs of q; line-delta quadrature against a brute-force two-dimensional integral to $5 \cdot 10^{-5}$ (grid limited); measure scaling Eq. (17) to 10^{-15}; quadrature converged to 10^{-9} with 128 points per locus.

- *Assembly*: action budget (transfer + tail exchange) closed to $3 \cdot 10^{-15}$; rotation equivariance, isotropy and cubic homogeneity at machine precision; discrete self-similarity Eq. (18) to 10^{-8}; trivial-filter insensitivity 1.2%.

- *Equilibria*: constant A null at $2 \cdot 10^{-19}$ of the one-sided (gross) rate; Rayleigh–Jeans null at $7 \cdot 10^{-5}$ with $O(h^2)$ convergence; near-vanishing transfer for the Zakharov–Filonenko spectrum $E \propto \nu^{-4}$.

- *Magnitude and shape*: mean JONSWAP transfer with the classic $(+, -, +)$ lobe structure; peak magnitude within a factor 0.8 of the WAM DIA it was calibrated against, ruling out any residual power of 2π or g in the normalisation chain.

- *Independent exact code*: on identical input, and with the default interaction filtering of that code disabled, the transfer lobes agree with the exact-NL code of van Vledder (WAVEWATCH III / SWAN) at the level of a few percent (positive lobe ratio 0.96, negative lobe ratio 1.02 in the reference case), with the independent result bracketed between the direct and conservative modes, as expected for consistent discretisations of the same integral.

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